Google Cloud H2S HACKATHONS

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Techno Crackers**

b. Team leader name: **Venket Raj S** (Team: Venket Raj S · Malik Chandra Pandian · Arun Veerappan · Nagaraj)

c. Problem Statement: **BFSI: Intelligent Risk, Fraud & Financial Experiences**

CatSight — Event-aware contract intelligence for catastrophe reinsurance: After a typhoon or earthquake, a reinsurer cannot say quickly or defensibly how much it owes because the answer depends on contract clauses buried in PDF treaty wordings.

Brief about the idea

CatSight tells a reinsurer in **about 15 seconds** after a typhoon or earthquake which treaties respond, what each layer pays, and which contract clauses could change that number — with every quote machine-verified against the source PDF page.

Real-world anchor — Typhoon Jebi (Japan, Sept 2018):

- **857,284 claims** and **¥1,067.8bn paid** — the largest ever for a Japanese typhoon GIAJ
- First modelled range was **$3–5.5bn**, arriving ~10 days after landfall Moody's RMS
- Final insured loss grew to **~$13bn** (~3× the early estimate), driven by storm surge & overlapping claims Swiss Re

Why it matters: While industry numbers take weeks, reinsurance teams spend **30–60 analyst-hours (2–4 business days)** assembling spreadsheets and re-reading 25+ exposed PDF treaty wordings by hand.

Source / Workflow	Arrives	Scope & Limitation
Modeller event response	**4–20 days**	Market range; ignores your PDF wording
PERILS (Japan typhoon)	**6 weeks**	Industry loss index, not treaty-level
Verisk PCS Japan	**up to 90 days**	Market-wide index for events ≥ $2bn
Manual in-house review	**2–4 days**	Error-prone spreadsheets; missed endorsements
CatSight (Lion Re's book)	**0.7 s math · ~10–12 s full AI check**	**Event × exposure × layers × verified PDF wording**

Replays real public hazard footprints (NOAA IBTrACS, USGS ShakeMap) over a synthetic book of **24 treaties written for 8 fictional insurers** in Japan, the Philippines, and Taiwan.

Opportunities

- **How different is it from any of the other existing ideas?**

  • **CatSight is the join:** Cat-model event response (Moody's RMS, Verisk) computes hazard footprints and modelled loss on pre-coded terms, stopping at the PDF wording. Document-AI platforms extract clauses at underwriting time, stopping before the storm track and layer math. Generic LLM chat over PDFs hallucinates arithmetic and lacks an audit trail.
  • **Complementary by design:** CatSight is not a cat model; a licensed vendor footprint can plug directly into our typed hazard interface.

- **How will it be able to solve the problem?**

  • **Deterministic Python loss engine (`Decimal`, half-up, 874 automated tests):** Computes regional gross loss, Cat XL layer erosion, burn %, reinstatements, reinstatement premium (RIP), and Quota Share cessions in **0.7–1.0 s** — the LLM never does arithmetic.
  • **RAG over Firestore vector search + 2-Stage Verifier Loop:** Retrieves governing clauses per treaty (252 vectors, 768-d `gemini-embedding-001`) and verifies every quote via deterministic normalised/fragment matching + a Gemini 3.8 Flash semantic support check (with 1 revision retry before withholding).

- **USP of the proposed solution**

  • **Adversarial Clause Courtroom:** Two blind parallel AI counsel argue opposing readings of ambiguous clauses (e.g., T-004 storm-surge exclusion) from verified quotes while the engine prices both readings per party (Insurer, All Reinsurers, Lion Re).
  • **Scanned Endorsement Detective:** Transcribes rasterised, stamped PDF pages with no text layer (T-004 p.7: 168h overrides 72h) and badges them `transcribed`.
  • **Community & APAC Impact:** 91% of APAC's 2023 cat losses ($65bn) were uninsured (MAS), and 48% of insurers license no cat model (Aon 2025). Faster, less-disputed reinsurance settlements help primary insurers pay households and SMEs sooner.

List of features offered by the solution

Feature	What it gives the catastrophe analyst	Verification & Trust Guardrail
1. Treaty Matching & Layer Waterfall	Matches all 24 treaties on peril ∩ territory ∩ period (with exclusion reasons for the other 11); computes per-layer ceded, burn %, reinstatements, RIP, and 3-party net positions (Insurer, All Reinsurers, Lion Re).	Deterministic `Decimal` engine; 100% match against hand-checked worked examples.
2. "How was this calculated?"	Step-by-step audit drawer per treaty: regional peak wind/MMI, vulnerability curve damage %, insured value × damage % = gross loss, and exact layer formula with citations.	Every intermediate number is inspectable and reproducible.
3. Cited Wording Flags & Endorsement Detective	Flags clauses that bite on *this* event (e.g., T-004 storm-surge exclusion on Jebi's 329 cm Osaka tide) and transcribes scanned image endorsements (T-004 p.7: 168h replaces 72h).	2-stage verifier loop; unverified flags visibly withheld (`n verified · k withheld`); scanned pages badged `transcribed` .
4. Clause Courtroom	Parallel ADK agents (`cedent_counsel` ‖ `reinsurer_counsel`) argue opposing readings from verified quotes (with 1 revision retry & `...` fragment matching); engine prices both readings; Arbiter summarises without deciding.	Issue-spotting only; human analyst records the decision with an audit timestamp.
5. Bounded What-if & Grounded Q&A	Interactive hours-clause scenario pricing (clearly labelled HYPOTHETICAL on Jebi's 9-hour window) and live follow-up Q&A over treaty clauses.	Typed, bounded tool schema (0–50% share); every figure in AI prose is checked against engine output.
6. Instant Replay, 9-Stop Tour & Word Export	6-second zero-cost replay of a recorded real run, live SSE streaming (~15 s), colour-blind-safe themes, keyboard navigation, and Word (`.docx`) & Markdown memos.	Every export carries the methodology version, verification counts, and mandatory disclaimers.

Process flow diagram or Use-case diagram

Workflow 1: Live Event Analysis Graph (ADK 2.11 `Workflow`)

1. event_intel
Loads NOAA IBTrACS / USGS ShakeMap footprint, event window (`start` , `end` , `duration_h`) & JMA facts

▼

2. exposure_analyst (NO LLM — Pure Python Engine)
Matches 24 treaties → regional gross loss → Cat XL / QS waterfall → emits `result` SSE frame in ~0.7–1.0 s

▼

3. wording_checker (Gemini 3.5 Flash-Lite)
Firestore 768-d vector search per treaty (`treaty_id` pre-filter) → candidate flags with page, clause & quote

▼

4. verifier ⇄ reviser (Bounded 1-Retry Loop)
Stage 1: deterministic normalised + `...` fragment quote match · Stage 2: Gemini 3.8 Flash semantic support check → 1 revision retry on failure → withhold if still unverified

▼

5. reporter (Gemini 3.5 Flash-Lite + Python Number Gate)
Drafts executive memo from verified facts; Python checks every number against engine output (0 untraceable figures)

Workflow 2: On-Demand Clause Courtroom (Parallel Fan-Out / Join)

1. brief
Selects top verified HIGH/MEDIUM exclusion flag + loads Clause 2 (Business Covered) & exclusion clauses

▼

2a. cedent_counsel
Gemini 3.5 Flash-Lite · argues insurer reading + excluded share (0–50%)

2b. reinsurer_counsel
Gemini 3.5 Flash-Lite · argues reinsurer reading + excluded share (0–50%)

▼ JoinNode + Quote & Support Check (with 1 revision retry per side)

3. price (Deterministic Python Scenario Engine)
Prices both verified readings through the treaty layers; computes `money_at_stake` per party by subtraction

▼

4. arbiter (Gemini 3.8 Flash) → Human Analyst Decision
Summarises both arguments without deciding; analyst records position (`cedent` / `reinsurer` / `unresolved`)

Wireframes/Mock diagrams of the proposed solution (optional)

Five-Step Progressive Analyst Interface Wireframe

Sticky Summary Bar (Always Visible)
Gross Claims ($733.04M) · Reinsurers Pay ($490.92M) · Lion Re Net ($70.36M) · Verified Flags · Elapsed · Method v1.0

Step 1 · Event
Track / ShakeMap + JMA facts + Instant Replay (6 s) or Run Live (~15 s)

Step 2 · Who's Hit
MapLibre choropleth map of 34 prefectures + accessible exposure table

Step 3 · What It Costs
3-party bar + 13 responding treaties + expandable "How was this calculated?" layer formulas

Step 4 · Clauses That Matter
Verified wording flags + Side-by-side PDF/Clause Drawer + Parallel Clause Courtroom + What-if

Step 5 · Decide & Export
Analyst position selector + figure-checked Executive Report + Word (`.docx`) / Markdown export

Trust Architecture: 2-Stage Citation Verification Funnel

Candidate Wording Flag / Counsel Argument
Model cites `treaty_id` , `page` , `clause_no` , and verbatim `quote`

▼ Stage 1: Deterministic Python Quote Match

Word-Aligned Normalised & Ellipsis (`...`) Matcher
Checks quote against stored Firestore chunk; tags scanned pages as `transcribed` (40/40 corrupted quotes caught in unit suite)

▼ Stage 2: Semantic Support Check (Gemini 3.8 Flash)

Entailment & Event-Window Verification
Verifies quote + event window (`start` , `end` , `duration_h`) + JMA facts directly support the flag

▼ Fail → 1 Revision Retry with Feedback → Still Fails?

Visibly Withheld (Never Shown as a Finding)
UI displays `n of n displayed citations verified · k withheld` with exact reason

Architecture diagram of the proposed solution

Analyst / Judge Browser (React 18 + TypeScript + MapLibre GL)
Same-origin HTTPS · REST + POST Server-Sent Events (`fetch` ReadableStream with 15 s heartbeat)

▼

Google Cloud Run — `catsight-api` (us-central1)
Single container serves React UI + FastAPI SSE API · Google ADK 2.11 `Workflow` graphs · Deterministic Python `Decimal` Loss Engine · Dedicated least-privilege SA (`catsight-run`) · `max-instances 3` · Per-IP hourly & daily cost caps

▼

Gemini Enterprise Agent Platform
(formerly Vertex AI, global)
- Gemini 3.5 Flash-Lite
- Gemini 3.8 Flash
- gemini-embedding-001

Cloud Firestore (Native)
us-central1
- Treaty book & events
- 252 clause vectors (768-d)
- Saved runs & positions

Cloud Storage
us-central1
- Synthetic treaty PDFs
- Stamped endorsements

▼

Cloud Logging · Cloud Trace (OpenTelemetry) · Cloud Monitoring · Log Analytics SQL
Structured JSON logs per stage · Guardrail metrics (`citation_withheld` , `counsel_dropped` , `unsupported_figures`) · Live Service Dashboard

Key Architectural Decisions

- **Single-origin deployment:** React UI and FastAPI SSE stream from one Cloud Run service — zero CORS issues or proxy buffering timeouts.
- **Zero API keys in code or env:** Application Default Credentials via dedicated `catsight-run` IAM service account.
- **Pre-embedded standard queries:** Ship in `query_embeddings.jsonl` so cold starts make zero embedding calls during an analysis; interactive `/ask` queries bypass batch rate-limit sleeps.
- **Spoof-resistant rate limiting:** Reads right-most hop of `X-Forwarded-For` appended by Cloud Run's load balancer; falls back gracefully to recorded replay if cap is reached.

Technologies to be used in the solution

Technology / Service	Component	Why it was chosen & role in CatSight
Gemini 3.5 Flash-Lite Gemini Enterprise Agent Platform (Vertex AI)	Agent generation & OCR	Fast, low-cost structured output (`response_schema` , `thinking_level=LOW`) for wording checks, parallel Courtroom counsel, report prose, and scanned-page transcription.
Gemini 3.8 Flash Gemini Enterprise Agent Platform (Vertex AI)	Verification & Arbiter	Stronger reasoning model used as the semantic support judge in the Verifier Loop, Counsel Quote Checker, and neutral Courtroom Arbiter.
Google ADK 2.11 `google-genai` + ADK Workflow	Multi-agent orchestration	Graph `Workflow` with bounded retry loops (`verifier ⇄ reviser`), parallel fan-out/fan-in (`JoinNode` for blind counsel), and OpenTelemetry Cloud Trace spans.
Cloud Firestore + Vector Search `gemini-embedding-001` (768-d)	Operational & vector store	Stores the 24-treaty book, 252 clause vectors with equality pre-filtering on `treaty_id` (186/186 governing clauses retrieved), ingestion audit trail, and runs.
Google Cloud Run & Cloud Storage	Serverless runtime & PDFs	Scale-to-zero container serving FastAPI SSE + React 18 UI; GCS stores treaty PDF wordings.
Cloud Logging, Trace & Monitoring	Production observability	Log-based metrics + Log Analytics SQL query (`observability_runs.sql`) powering the Cloud Monitoring dashboard.
Python 3.12 · FastAPI · React 18 · MapLibre	Core engine & frontend	Deterministic `Decimal` loss engine (874 pytest tests), python-docx memo generator, accessible TypeScript UI.

Estimated implementation cost (optional)

$0.010
p50 model cost per live analysis ($0.008–$0.013)

$0.004
p50 model cost per Clause Courtroom run

$0.000
Instant Replay cost (cached real SSE stream)

$0.003
Follow-up question (from token counts)

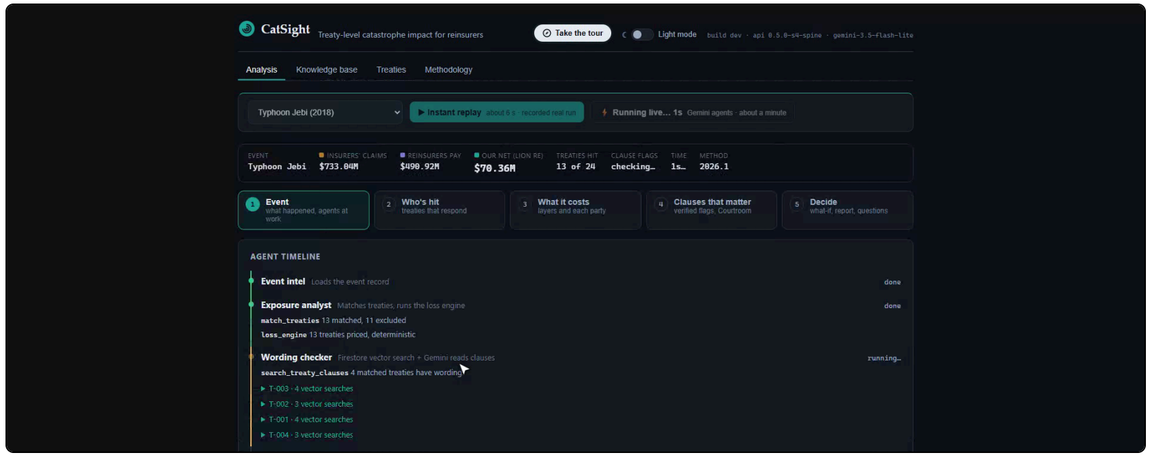
Measured Cloud & Token Breakdown

Workload	Model / Service	Measured Cost
One-time ingestion (8 PDF wordings, 252 chunks)	`gemini-embedding-001` + `3.5-flash-lite`	< **$0.08 total** (57 s runtime)
Live event analysis (13 responding treaties)	`3.5-flash-lite` + `3.8-flash` verifier	**$0.0094–$0.0102 p50**
Clause Courtroom (parallel counsel + arbiter)	2× `3.5-flash-lite` + `3.8-flash`	**$0.0040 p50**
Cloud Run + Firestore (scale-to-zero default)	`min-instances 0`, `max-instances 3`	**~$0 idle** · pennies per active hour

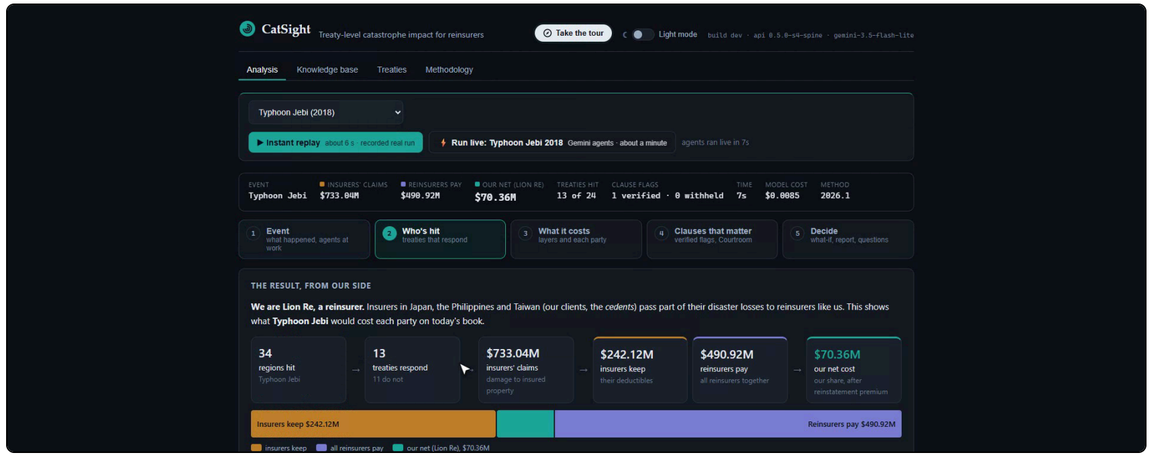
Annual Run-Rate vs. Manual Analyst Effort

- **Mid-size reinsurer workload:** 8 major events + 20 smaller checks/year (≈ 500 live analyses & Courtroom runs/year) → **~$7/year in Gemini token cost** + ~$25–$40/month Cloud Run/Firestore when active.
- **Estimated manual effort saved:** 30–60 analyst-hours per event reduced to 3.5–8 hours of review → **165–675 analyst-hours/year** (est. $12k–$100k/year in specialist time; labelled *simulated estimate*).
- **Built-in cost guards:** 20 live runs/hour per visitor, 300/day service cap, automatic fallback to Instant Replay, and Cloud Billing budget alert.

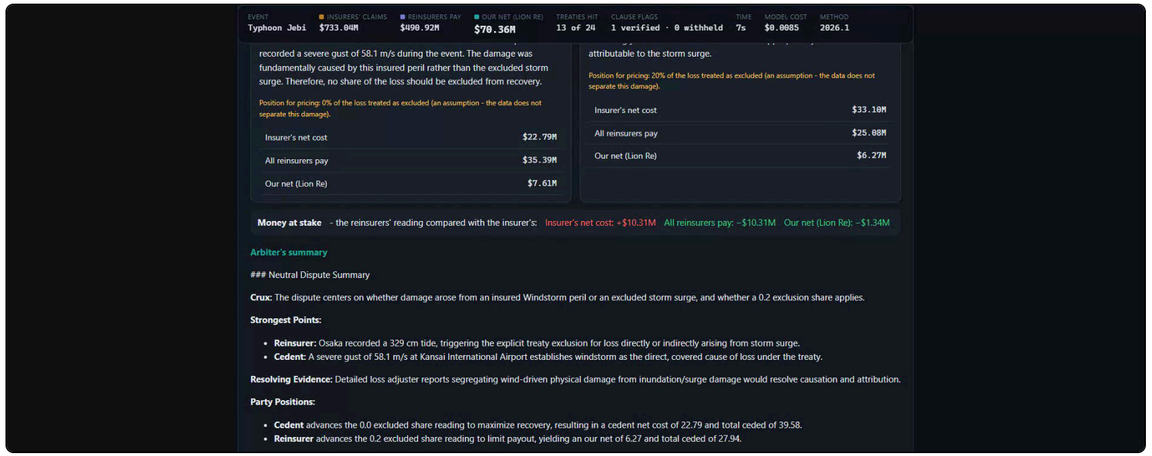
Snapshots of the prototype



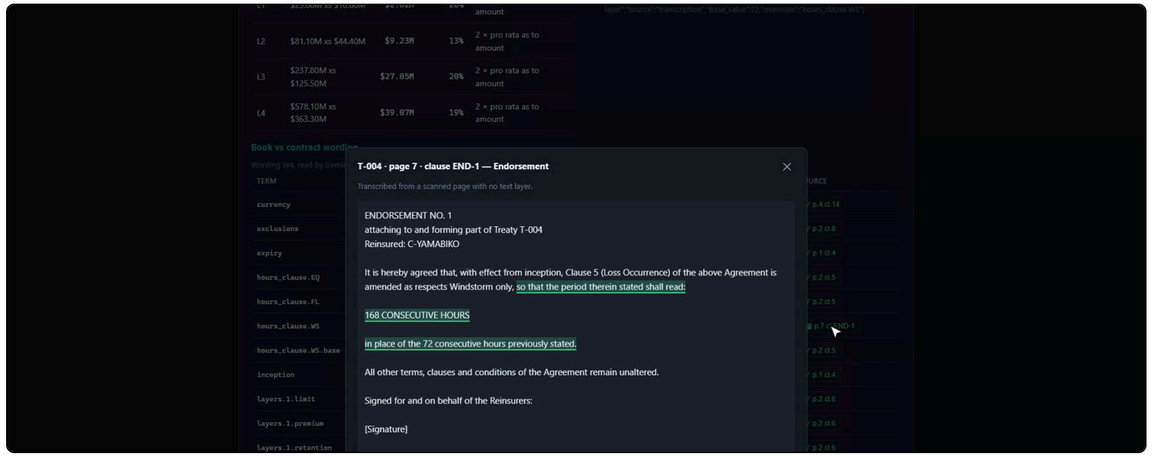
1. Live Agent Streaming: ADK 2.11 stages stream over SSE; deterministic loss figures arrive in ~0.7–1.0 s while Gemini retrieves & verifies treaty clauses.



2. Three-Party Loss Waterfall: 13 of 24 treaties respond on Typhoon Jebi ($733.04M claims → $242.12M insurer retention, $490.92M reinsurers, $70.36M Lion Re net).



3. Clause Courtroom (Jebi T-004 Storm-Surge Exclusion): Parallel counsel argue 0% vs 20% surge share; engine prices both (Ceded −$11.64M, Insurer +$10.31M, Reinsurers −$10.31M, Lion Re −$1.34M).



4. Scanned Endorsement Detective (T-004 p.7): Gemini transcribes a rasterised stamped endorsement with no text layer (168 hours replaces 72 hours) and badges it `transcribed`.

Prototype Performance report/Benchmarking

0.7–1.0 s
Time to money on screen (p50 across two 5-run live benchmarks)

9.8–12.1 s
Complete live analysis with verified flags (p50; range 8.0–31.2 s)

99.5%
RAG treaty-term extraction (185/186 fields in 57 s vs ~25 min whole-doc)

874
Automated pytest tests (861 offline unit + 13 live GCP; CI on GitHub)

Benchmark / Evaluation (`docs/RESULTS.md`)	Measured Result	Methodology & Honest Caveat
Loss engine vs. hand-checked worked example	**100% exact match**	`Decimal` half-up; verified on every Cat XL layer, reinstatement, RIP, and Quota Share formula.
Governing clause retrieval & citation audit	**186/186 retrieved · 185/186 verified**	Firestore 768-d vector search; 184 text-layer verified + 1 scanned endorsement transcribed; 1 unverified withheld.
Corrupted quote rejection (Verifier unit suite)	**40/40 rejected (0 false withholds)**	Normalised + word-aligned `...` ellipsis fragment matcher + Gemini 3.8 Flash semantic check.
Untraceable numbers in AI-written reports	**0 across all live runs**	Deterministic Python post-check compares every figure in reporter/arbiter prose against engine output.
Honest limitations & unmeasured items	Extraction prompt was tuned on the 8 synthetic wordings (99.5% is an upper bound; blind set is next). Manual effort savings (30–60h → 3.5–8h) are labelled *simulated estimates* pending a timed practitioner study.	

Additional Details/Future Development (if any)

The Money Moment: Why Per-Party Pricing Matters

1. Live Jebi Courtroom (T-004 Storm-Surge Exclusion, p.2 cl.8):
At a 20% surge share (recorded live run), total ceded drops by **−$11.64M** ($39.58M → $27.94M), the insurer's net cost rises by **+$10.31M**, all reinsurers save **−$10.31M**, and Lion Re saves **−$1.34M** (at 30% share: Lion Re saves **−$2.44M**).

2. Worked Example Hours-Clause Asymmetry (What-if):
On a $54.4M loss across L1 (20 xs 10, Lion Re 25%) and L2 (30 xs 30, Lion Re 10%), splitting 1 event into 2 ($35.0M + $19.4M) cuts market ceded recoveries by **−22.5% (−$10.0M)**, but **increases Lion Re's net loss by +7.4% (+$0.51M, $6.82M → $7.33M)** because the split shifts burden down into Layer 1 where Lion Re's share is concentrated.

Team Techno Crackers — Roles

- **Venket Raj S (Lead):** Architecture, Cloud Run / Firestore / IAM deployment & observability
- **Malik Chandra Pandian:** Reinsurance domain modelling, `Decimal` loss engine & worked examples
- **Arun Veerappan:** ADK 2.11 workflows, RAG extraction, citation verifier & Clause Courtroom
- **Nagaraj:** React 18 + TypeScript + MapLibre UI, 5-step analyst workflow, tour & exports

Future Development Roadmap (Designed / Investigated)

- **Real multi-shock hours-clause dispute (2024 Hualien Earthquake, Taiwan):** Investigated in repo — its real aftershock cluster ~20 days later creates a genuine second occurrence worth ~$11–15M per treaty under the hours clause.
- **Calibrated per-country vulnerability curves:** Adding insurance-penetration & insured-loss wind curves for the Philippines (bringing Typhoon Haiyan into the live selector).
- **Vendor hazard footprint adapter:** Plugging licensed Moody's RMS or Verisk footprints behind our typed hazard interface.
- **Authenticated treaty upload & BigQuery exposure:** Scaling from 24 treaties to enterprise treaty packs.
- **Live USGS / GDACS Pub/Sub intake & A2A/MCP endpoint:** Automated first-view triggers on disaster alerts and broker-agent interoperability.

Provide links to your:

- **GitHub Public Repository**
  `https://github.com/venketraj/catsight-ai-buildercup-2026`
  Apache-2.0 licence · 874 automated pytest tests · GitHub Actions CI · Full architecture, per-field citation audit & assumptions in `docs/`.

- **Demo Video Link (3 Minutes)**
  [VIDEO_LINK_PLACEHOLDER — Insert Public YouTube / Demo Video URL (< 3:00)]
  3-minute walkthrough of the reinsurance problem, live ~15 s Typhoon Jebi analysis, scanned endorsement transcription, and Clause Courtroom.

- **Final Product Link**
  `https://catsight-api-342162443546.us-central1.run.app`
  Live on Google Cloud Run (us-central1) · No sign-up required · Click **Instant replay** (~6 s) or **Run live** (~15 s) · 9-stop guided tour included.

Disclaimers & Data Attribution: Indicative first view on synthetic data. Not a catastrophe-model output. Not underwriting or claims advice. Issue-spotting only — not legal advice. Jebi, Hagibis, Haiyan and Noto were real events in which people died. Portfolio, insurers and treaty wordings are 100% synthetic; hazard data is public (NOAA IBTrACS v04r01, USGS ShakeMap, Natural Earth).

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

